MAIS PAC

IPhO 04.02: Vectors and Complex Numbers · Lecture + Problems

Vectors and Complex Numbers

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

5 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. Why vectors exist
2. Components and the algebra of vectors
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1. Why vectors exist

A quantity that is not a single number needs a direction as well as a size. Force, velocity, momentum, electric field and angular momentum are all vectors; mass, energy, charge and time are not. The whole point of treating them as vectors is that they add by the parallelogram rule, not by simple arithmetic.

The single most useful sentence in this topic: **choose axes parallel and perpendicular to the motion or to the unknown force, resolve everything onto them, and then treat each direction as an independent one-dimensional problem.** Almost every "hard" mechanics problem becomes trivial once the axes are chosen well.

2. Components and the algebra of vectors

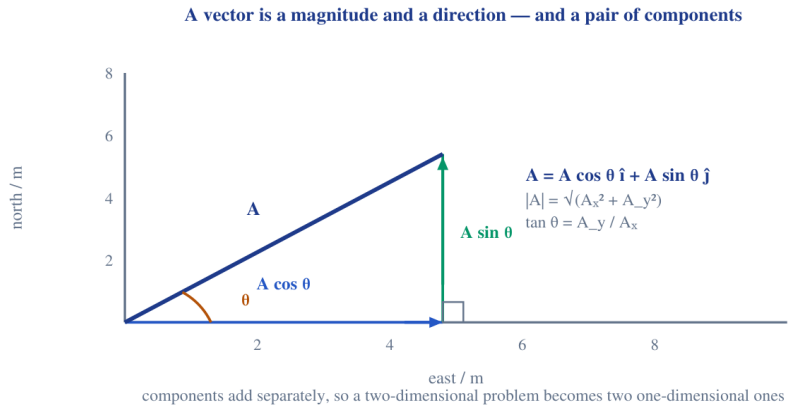


Figure 1. Resolution and composition are the same operation read in two directions. Because components along perpendicular axes never mix, every vector problem splits into independent one-dimensional problems.

For a vector of magnitude A at angle θ to the x-axis, the components are A_x = Acos θ and A_y = Asin θ, with

$$A = \sqrt{(A_x^2+A_y^2)},\ \tan\theta = (A_y)/(A_x). \tag{1}$$

In components everything is linear:

$$\{A\}+\{B\} = (A_x+B_x,;\ A_y+B_y),\ k\{A\} = (kA_x,;\ kA_y). \tag{2}$$

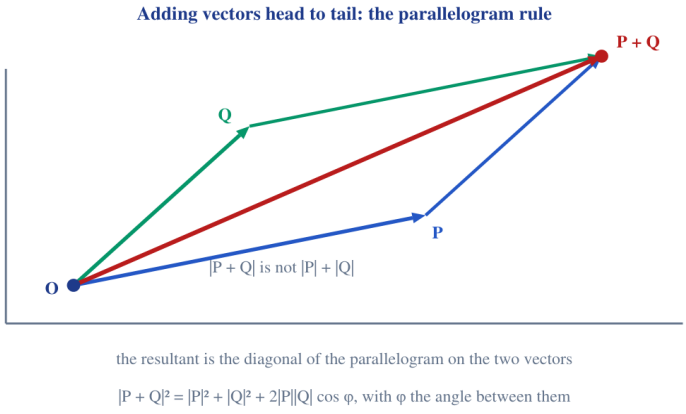


Figure 2. Only when the vectors are parallel does the sum have length equal to the sum of the lengths. The angle between them decides everything, and the cosine rule quantifies it.

The magnitude of the sum follows from the cosine rule, with φ the angle **between** the vectors:

$$|\{A\}+\{B\}|^2 = A^2+B^2+2AB\cos\varphi. \tag{3}$$

Two special cases are worth remembering: parallel vectors (φ = 0) add directly, antiparallel ones (φ = 180°) subtract, and perpendicular ones give |{A}+{B}| = √(A²+B²) and no cross-talk between components.

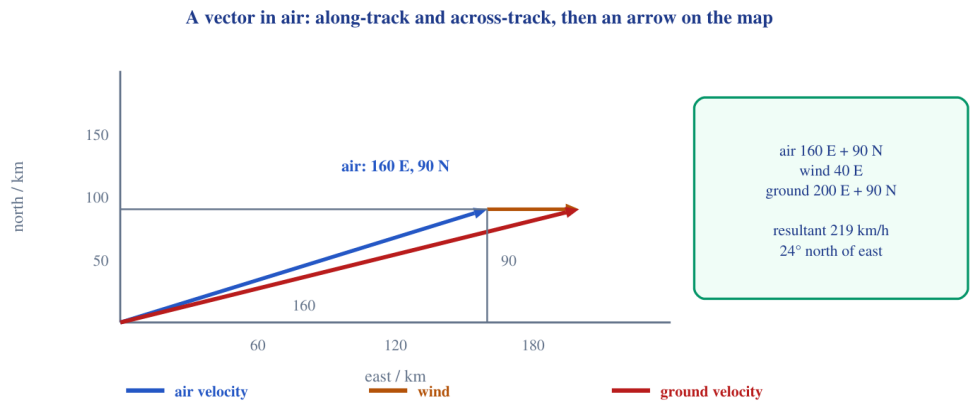


Figure 3. Components make the addition mechanical: add the east components, add the north components, then recombine. The angle of the resultant comes from its two components, not from the two lengths.

3. The dot and cross products

Two different products appear constantly, and they answer two different questions:

product	definition	answer	typical use
scalar {A}·{B}	ABcos ϕ = A_xB_x + A_yB_y + A_zB_z	a number	work, flux, component along a direction
vector {A}×{B}	ABsin ϕ, direction by the right-hand rule	a vector	torque, magnetic force, angular momentum

The dot product is the "how much lies along" product. Its cross product partner is the "how much is perpendicular" product, with magnitude |{A}×{B}| = ABsin φ — the area of the parallelogram the two vectors span, which is why the cross product appears whenever an area, a torque, or a rotation is involved.

Both products are invariant under a rotation of the axes. That is their practical value: choose whatever axes make the components easy and the physics does not change.

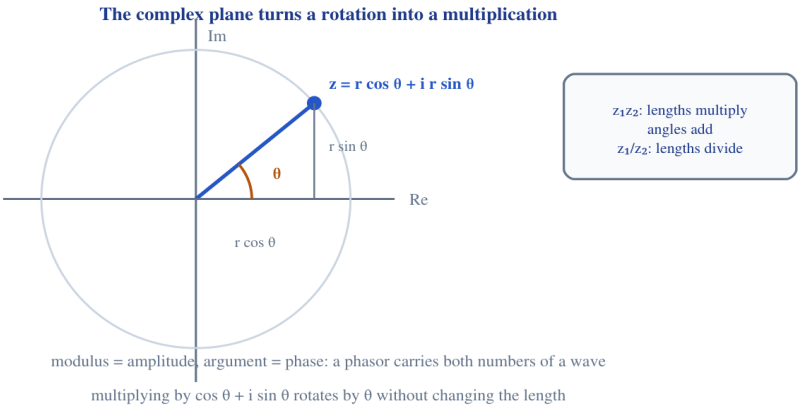


Figure 4. Phasors are the practical form of complex numbers: adding waves means adding their phasors, and a phase shift becomes a rotation. In AC circuit theory this turns trigonometry into bookkeeping.

4. Complex numbers: the same plane, with multiplication

A complex number z = x + iy is a vector in a plane, but a richer one, because two complex numbers can be multiplied. Writing z = r(cos θ + isin θ) with r = |z| and θ = z,

$$z_1z_2 = r_1r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]. \quad (4)$$

So multiplication multiplies the lengths and adds the angles. That makes complex numbers the natural language for anything that rotates: AC circuits, phasors, wave superposition, and the solution of linear differential equations.

A few consequences that carry straight into physics:

- $|z|^2 = zz^*$ is the square of the amplitude, so the intensity of a wave is the squared modulus of its phasor.
- $e^{i\theta} = \cos\theta + i\sin\theta$ turns a phase shift into a multiplication.
- The nth roots of unity, $e^{2\pi ik/n}$, are evenly spaced on the unit circle — the mathematics behind symmetry arguments and Fourier analysis.

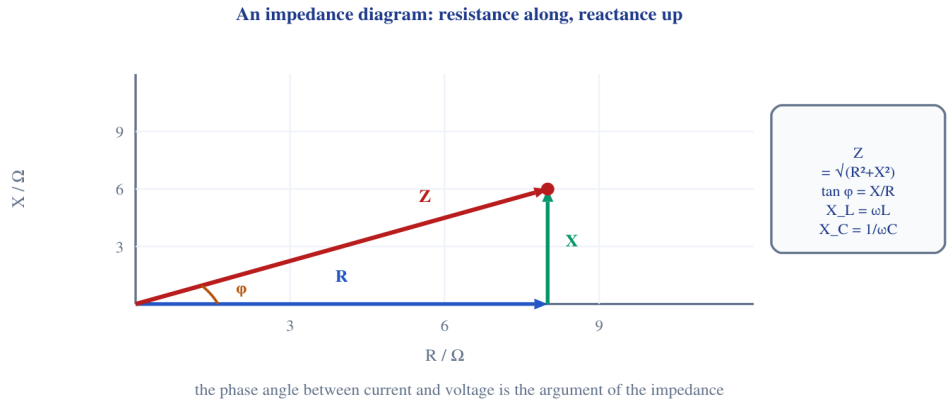


Figure 5. Impedance behaves exactly like a two-dimensional vector, except that the vertical axis is reactance. Because the two axes are perpendicular, the magnitudes do not simply add: the impedance magnitude is the hypotenuse of R and X.

For an AC circuit with resistance R, inductive reactance $X_L = \omega L$ and capacitive reactance $X_C = 1/\omega C$, the impedance is $Z = R + i(X_L - X_C)$. The current lags or leads the voltage by Z, and the magnitude is $|Z| = \sqrt{(R^2+X^2)}$. Every "two components at right angles" formula in physics has this shape for the same reason: the components are independent, so the magnitudes combine in quadrature.

Interactive experiment: Vector addition and resolution — open the online handout to use it.

The vector method in five steps. Draw the diagram and label every vector → choose perpendicular axes, ideally along the unknown or along the motion → resolve each vector, writing components with correct signs → write one equation per axis (equilibrium means each sum is zero; motion means F = ma on each axis) → recombine only at the end, and check the answer in a limiting case such as a right angle or a zero vector.

5. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Components and a resultant. A force of 50 {N} acts at 30° above the horizontal, and a second force of 40 {N} acts at 120° above the horizontal. (a) Resolve both into horizontal and vertical components. (b) Find the magnitude and direction of the resultant. (c) Explain why the resultant is smaller than 90 {N}.

Worked solution →

[1] Problem 2 — Straight-line motion in two dimensions. A boat can travel at 4.0 {m,s^(-1)} in still water and is aimed directly across a river 60 {m} wide flowing at 3.0 {m,s^(-1)}. (a) Find the resultant speed relative to the bank. (b) Find how far downstream the boat lands. (c) Find the heading needed to land directly opposite.

Worked solution →

[2] Problem 3 — Dot and cross products. {A} = 3{i} + 4{j} and {B} = -1{i} + 2{j}. (a) Find {A}·{B}. (b) Find the angle between them. (c) Find |{A}×{B}| and verify that |{A}×{B}| = ABsin φ.

Worked solution →

[2] Problem 4 — Equilibrium with three forces. A lamp of weight 30 {N} hangs from two wires making 40° and 60° with the horizontal. (a) Write the equilibrium equations. (b) Find the tension in each wire. (c) Explain what happens as the wires are pulled towards the horizontal.

Worked solution →

[2] Problem 5 — A phasor calculation. A voltage V_0cos ω t drives a series R–L circuit with R = 30 Ω and X_L = 40 Ω. (a) Write the impedance as a complex number. (b) Find |Z| and the phase angle. (c) Find the current amplitude if V_0 = 100 {V}, and state whether the current leads or lags.

Worked solution →

[2] Problem 6 — Superposition with phasors. Two coherent waves of amplitude A arrive with a phase difference of 60°. (a) Add them as phasors. (b) Find the resultant amplitude. (c) Find the intensity relative to a single wave.

Worked solution →

[2] Problem 7 — Work as a dot product. A block is dragged 5.0 {m} across a floor by a force of 20 {N} directed 35° above the horizontal. (a) Find the work done by the applied force. (b) Find the work done by friction if μ = 0.25 and m = 4.0 {kg}. (c) Explain why the applied force does less work than 100 {J}.

Worked solution →

[3] Problem 8 — A relative-velocity overtaking problem. Car A travels east at 20 {m,s^(-1)} and car B travels north at 15 {m,s^(-1)}. (a) Find the velocity of B relative to A. (b) Find its magnitude and direction. (c) Explain how the answer would look to a passenger in B.

Worked solution →

[3] Problem 9 — Minimum force using components. A block of weight W rests on a plane inclined at θ. (a) Resolve the weight along and perpendicular to the plane. (b) Find the minimum force needed to keep the block from sliding, and in what direction it must be applied. (c) Show that for μ = 0 the answer reduces to the familiar result.

Worked solution →

[3] Problem 10 — Complex numbers and roots of unity.

Show that $z = e^{i\theta}$ satisfies $|z| = 1$ for all real θ. (b) Find the three cube roots of 1 and plot them. (c) Show that the three roots add to zero, and explain what that means for three equal forces directed at 120° to one another.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

Paul's Online Math Notes

Calculus, series and approximation practice at exactly the level the syllabus expects.